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IGXLAdminDefTopic

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About IG-XL Administration

[About IG-XL Administration](#)

This section includes topics relating to IG-XL administration. The following topics are available:

- | | |
|---|---|
| • | IG-XL Licensing |
| • | Switching IG-XL Versions |
| • | Installing Custom DLLs |
| • | Virus Protection |
| • | Validating Your Operating Environment |
| • | Remote Test System Connections |

adCustomDLL



Installing Custom DLLs

Installing Custom DLLs

A Dynamic Link Library (DLL) is a collection of procedures and functions that resides in one file (with the file type .dll) and that is available to any Microsoft Windows program that knows how to call it. Windows loads the routines in the file only when a program requires them, and it can unload them if they are not in use to save memory. To make routines in the DLL available to programs, you must register the DLL so that Windows can manage it.

Most DLLs are part of a larger package, and the package's Setup installs and registers them. This is true for all DLLs needed to run IG-XL. If you write your own DLLs (for example, to provide a specific datalog format), or if you need to install a custom application or package such as a hardware driver, you may need to register the DLL yourself. (Handler/prober drivers often require that you register them yourself.) Note that not all DLLs require registration, so make sure that your custom DLL does require it before you proceed.

Installing a DLL

You can install a custom DLL in two ways:

- If you have the source code and the system on which you want to install the DLL has Visual Basic, compile the source and create the DLL. This automatically registers it.
- If you have the prepared .dll, copy the file to an appropriate place on your hard disk. If you need to register the DLL, use the regsvr32 utility provided with Windows. This utility registers a DLL and confirms its registration. From a command prompt, type:

`regsvr32 <DLL-path-and-file-name>`

For a description of regsvr32 syntax, type regsvr32 and press Enter at a command prompt.

You must have Administrator privileges to use the regsvr32 command.

Removing a DLL

You can unregister a DLL to make its procedures and functions unavailable to Windows programs. You may want to do this if:

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|--|---|
| | <ul style="list-style-type: none">You want to replace a DLL with a newer version |
| | <ul style="list-style-type: none">Your system no longer contains the hardware that the DLL supports |
| | <ul style="list-style-type: none">You want to remove the functionality that the DLL provides |

Make sure that you do not unregister a DLL that Windows or IG-XL requires. Doing so can cause the operating system or certain programs to stop functioning properly. In general, do not unregister a DLL that you did not register yourself.

To unregister a DLL, open a command prompt and type:
regsvr32 /u <DLL-path-and-file-name>

To minimize the risk of confusion when unregistering a DLL:

- | | |
|--|---|
| | <ul style="list-style-type: none">Before registering a new version of a custom DLL, first unregister the old version. |
| | <ul style="list-style-type: none">After you unregister a DLL, remove it from your computer. |

If you unregister a DLL that provides driver support for specific hardware, that hardware no longer appears in the EquipmentSelect list. However, it is not deleted from the tester.

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IG-XL Licensing

IG-XL Licensing

IG-XL requires a license to run, and some of its features require separate licenses. Teradyne’s Self Service Licensing (SSL) provides software license management for all IG-XL users. SSL is available for all versions of IG-XL on all IG-XL based platforms and is the method you use to obtain and administer IG-XL licenses.

Access to the Teradyne Self Service Licensing Portal is available at:
<https://teradyne.subscribenet.com>. To access the portal you must have access to your organization’s account. To obtain an account, contact your account administrator.
This section provides an overview of licenses for UltraFLEX test systems and includes the following topics:

- | | |
|---|--|
| • | Enabling Licensed Features |
| • | UltraFLEX Instrument Licensing |
| • | Available UltraFLEX Features |

For more information on licenses, contact your Teradyne sales or field representative.

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IG-XL Licensing : Enabling Licensed Features

Enabling Licensed Features

IG-XL allows you to enable features at various levels. This allows you to make efficient use of tester hardware without paying for capabilities that you do not need. For example, if a device does not require the full speed of the HSD subsystem, you can purchase a license for digital patterns at a lower speed.

Instruments that depend on the output of other instruments may require licenses for both. Check your tester configuration to make sure that you understand all such dependencies.

Products and features that require a license function only up to the level allowed by the license file. Each license file contains a list of available features and the levels at which you can use them.

When you try to use a feature not explicitly enabled in your license file, IG-XL behaves differently depending on your location:

- On a test system, your test program either fails validation or returns an error at test program load, depending on the type of license. For example, a missing DataTool license (that is, no license to run IG-XL at all) disables test program loads.
- When running on a simulator, IG-XL enables all features at the most advanced level. For information about customizing this behavior, see Setting Off-Line License Options.

To change your license options, contact your Teradyne sales or field representative.

The following topics are available:

- Determining a Test Program’s License Requirements
- Setting Off-Line License Options

For information on licensing specific IG-XL or instrument features, click [Related Topics](#):

For more information, see:

- | | |
|---|--|
| • | UltraFLEX Instrument Licensing |
| • | Available UltraFLEX Features |

	
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IG-XL Licensing : [Enabling Licensed Features](#) : Determining a Test Program’s License Requirements

Determining a Test Program’s License Requirements

When you develop a test program on a workstation, IG-XL enables all licenses at their most [advanced levels](#). (If your `SimulatedConfig.txt` has a `<LICENSE_ENABLERS>` section, however, licensing in that section is used instead. See [Setting Off-Line License Options](#).) When you run the test program on a test system, you must have a license for each feature that you use. To determine which features, if any, you need to request, you must know what capabilities your test program uses that require a license.

[On the first run after you load a test program](#), IG-XL creates the following files in the `\tester` folder of your current IG-XL version:

LicenseMappingFile.txt	This file lists the standard IG-XL license features on your computer (workstation or test system) and specifies which licenses were used by the test program. For a user-specified license (USL), this file only lists used USL features. It does not specify all available USL features. See LicenseMappingFile .
LicenseMapping_HSD.txt	This file provides license allocation information for memory depth, data rate, and digital channel availability. See: HSD1000 Software Licensing Requirements UltraPin800 Software Licensing Requirements UltraPin1600 Software Licensing Requirements

If you are running the test program online, IG-XL also creates the following file in your test file directory:

LicenseRequirements.txt	This file lists the minimum licenses required for your test program. See LicenseRequirements File .
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LicenseMappingFile

[LicenseMappingFile.txt](#) is generated the first time you run your test program. It lists:

- Standard IG-XL license features on your workstation or test system and which licenses were used by the test program
- User-specified license (USL) features used in the test program

For [system-wide licenses](#), [LicenseMappingFile.txt](#) reports only whether you are using a license. For example, if you have one tester-wide license that enables 48 VHFAC channels, IG-XL reports the license as in use even if your test program uses a single channel. This file also does not list required system-wide licenses that you do not have. Instead, IG-XL returns run-time errors for such missing licenses.

To determine which licenses your test program uses:

1. [Open your test program with license enforcement enabled:](#)
 - Run IG-XL on the test system, or
 - Create a <LICENSE_ENABLERS> section in your workstation's [SimulatedConfig.Txt](#) file (see [Setting Off-Line License Options](#)).
2. [Run the test program.](#)
3. [View](#) [LicenseMappingFile.txt](#) in the \tester folder.

The [license summary](#) in [LicenseMappingFile.txt](#) has the following sections:

License Mapping Feature Vs SlotNo

Describes the allocation of counted licenses. For each license at each enabled level, the SLOTS column lists which boards (identified by tester slots) use the license at the specified level.

In the following example, two SB6G boards (in slots 14 and 17) use the full vector rate of 6.4 Gbps, and one board (in slot 8) uses a slower vector rate of 3.2 Gbps:

#FEATURE	ENABLED LEVEL	SLOTS
HSS_Speed	6400	17,14
HSS_Speed	3200	8

IG-XL repeats the slot number if more than one license is at a particular enabled level. The slot number is shown as -1 for licenses that can span multiple slots, such as HSD_Speed, HSD_MemoryDepth and UltraVChannel.

Non-Enabled Feature License Availability

Lists software-only features. Your test program needs a license for each feature that has a Yes in the CURRENTLY IN USE? column.

Enabled Feature License Availability (Counted)

Lists the counted licenses available on your test system and how many of each feature license your test program requires. If you have a feature licensed at multiple levels (for example, you have licenses for multiple HSD data rates), each enabled level appears on a separate line. Each feature line includes the enabled level, the number of licenses your test program requires, and the number of unused licenses. If you do not have enough licenses to enable the resources your test program requires, the AVAILABLE LICENSES column shows the deficit as a negative number. IG-XL returns a run-time error if you run the test program without enough licenses.

Enabled Feature License Availability (Uncounted)

This section lists the node-locked licenses (and, if you are running on a workstation, floating licenses) available on your test system and whether your test program uses these licenses. Your test program needs a license for each feature that has a Yes in the CURRENTLY IN USE? column. Note that a license appears as in use even if your test program uses only a portion of that license. For example, if you have a license enabling use of all 48 possible BBAC channels but your test program uses only 12 channels, the license appears as in use.

Feature names appear alphabetically within each section.

The following is a sample UltraFLEX LicenseMappingFile.Txt:

#License Mapping Feature Vs SlotNo:

#FEATURE	ENABLED	LEVEL	SLOTS
HSD_MemoryDepth	8	-1	
HSD_Speed	200	-1	
NumSlots	1	-1,-1,-1,-1,-1,-1,-1,-1,-1,-1,-1,-1,-1	

#NonEnabled Feature License Availability:

#FEATURE	CURRENTLY IN USE?	USL
IG-XL_Bitmap	No	
IG-XL_DataTool_Jaguar	Yes	
IG-XL_ExChange	No	
IG-XL_PBM	No	
IG-XL_RA	No	
IG-XL_RAPlus	No	
IG-XL_SECSGEM	No	
IG-XL_TesterServiceDaemon	No	
IGXLConcurrentExecution	Yes	PPP
VX_HSD_FLEX	No	

#Enabled Feature License Availability (Counted):

#FEATURE	ENABLED	LEVEL	LICENSES IN USE	AVAILABLE LICENSES	USL
HSD_MemoryDepth	8	1	1	PPP	
HSD_MemoryDepth	8	0	1		
HSD_Speed	200	1	0	PPP	
HSD_Speed	50	0	2		

#Enabled Feature License Availability (Uncounted):

#FEATURE	ENABLED	LEVEL	CURRENTLY IN USE?	USL
BBACCap	4	Yes	PPP	
BBACCapBandwidth	3	Yes	PPP	
BitmapTool.Integrated	1	No		
BitmapTool.StandAlone	1	No		
WaferMap.Integrated	1	No		
WaferMap.StandAlone	1	No		

LicenseRequirements File

[LicenseRequirements.txt](#) is generated the first time you run your test program if you are running online. This file lists the minimum licenses required for your test program, including:

- Each licensed feature used by the test program

- The enabled level for each feature
- How many licenses each feature uses in the test program
- The version of the license for each feature

Including this file in your test program release package can help production facilities to configure testers with the correct licenses.

The following is an example `LicenseRequirements.txt` file.

```
#Minimum License Required By Job:      myTI245Program_UVI80_Day5.xls
#FEATURE NAME      ENABLED LEVEL  USED LICENSE COUNT  VERSION
12GHzReceiver      Yes          1          0.0
12GHzSource        Yes          1          0.0
1PortVNA           Yes          1          0.0
DUTLOSource        1           1          0.0
HSD_Speed          50          1          0.0
HSD1600_Channels   1           1          0.0
IG-XL_DataTool_Jaguar  Yes          1          8.1
LowPhaseNoiseDUTRefSource  1           1          0.0
NoiseSource        Yes          1          0.0
OnBoardDUALAWGSynthDR  1           2          0.0
UltraVIChannel     1           2          0.0
```

If you are running a test program online, you can also use the method `TheExec.Licenses.GenerateLicenseRequirementsFile` to generate this file at any time.

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IG-XL Licensing : Enabling Licensed Features : Setting Off-Line License Options

Setting Off-Line License Options

When working off-line on a workstation, the IG-XL simulator fully enables all features and capabilities. The <LICENSE_ENABLERS> section of the SimulatedConfig.txt file, if present, will make the off-line simulate the capabilities enabled. This allows you to verify that a tester has sufficient licenses to run your test program.

If the license section is not present, copy the entire <LICENSE_ENABLERS> section from the CurrentConfig.txt file on the tester you want to emulate to your workstation’s SimulatedConfig.txt file. CurrentConfig.txt is always stored in the \tester folder. If you performed a standard IG-XL installation, the files are in:

C:\Program Files (x86)\Teradyne\IG-XL\<version>\tester

where <version> changes according to the version of IG-XL you are running. SimulatedConfig.txt can exist in your local test program folder, the \tester folder, or the \bin folder:

C:\Program Files (x86)\Teradyne\IG-XL\<version>\bin

Following is a sample CurrentConfig.txt licensing section:

<LICENSE_ENABLERS>

#Off-line enabling levels

#FEATURE	ENABLED LEVEL	LICENSE COUNT	LEGAL ENABLING POINTS
HSD_MemoryDepth	8	1	256,128,96,64,32,16,8
HSD_Speed	200	1	1000,800,700,600,500,400, 300,200,100,50
VHFACapture	2	N/A	48,46,44,42,40,38,36,34, 32,30,28,26,24,22,20,18, 16,14,12,10,8,6,4,2,1,0
VHFACaptureSamplingRate	80	N/A	125,80
VHFACSource	2	N/A	48,46,44,42,40,38,36,34,

			32,30,28,26,24,22,20,18, 16,14,12,10,8,6,4,2,1,0
VHFACSourceSamplingRate	100	N/A	400,100
BBACCap	1	N/A	48,46,44,42,40,38,36,34, 32,30,28,26,24,22,20,18, 16,14,12,10,8,6,4,2,1,0
UltraCapture	3	1	75,40,3
UltraCapture	40	1	75,40,3
UltraCapture	75	1	75,40,3
</LICENSE_ENABLERS>			

where:

FEATURE	Name of the feature the license enables.
ENABLED LEVEL	Enabled feature level for this license.
LICENSE COUNT	Number of test system resources permitted to use this license simultaneously. "N/A" means the feature is node-locked, not counted.
LEGAL ENABLING POINTS	List of all allowable enabling points (for reference only).

You can replace the value in the ENABLED_LEVEL column with any of the values in the LEGAL_ENABLING_POINTS column. This can help you determine what licensing levels your test program requires.

If the <LICENSE_ENABLERS> section omits any required license, IG-XL uses the default behavior for that license. This allows you to manipulate the behavior of a subset of your licenses, including a single license, without requiring you to construct or copy the entire section.

If you modify SimulatedConfig.txt, you must restart DataTool or reload your test program for the changes to take effect.

For more information, see [UltraFLEX Instrument Licensing](#).

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IG-XL Licensing : UltraFLEX Instrument Licensing

UltraFLEX Instrument Licensing

The following IG-XL components require a valid license file:

- | | |
|---|---------------------|
| • | IG-XL DataTool |
| • | Bitmap Tool |
| • | Redundancy Analysis |
| • | RAPlus |
| • | Production Bit Map |
| • | SECS/GEM |
| • | ESA Toolkit |

The following table lists licensing requirements for each UltraFLEX instrument:

IG-XL Licensing Summary		
Instrument	Licensed Features	Comments

AWG6G and AWG6G Plus	None	Some license files may contain obsolete license feature lines for AWG. However, a license for AWG is not required.
BBAC Capture	- Number of channels - Bandwidth	See BBAC Capture Software Licensing Requirements.
BBAC Source	- Number of channels - Bandwidth	See BBAC Source Software Licensing Requirements.
DataTool	Enable	IG-XL does not start if this license is not present. The enable level of this license must be equal to or higher than the IG-XL version being used.
DC30	None	Use of DC30 does not require a license.
DC75	None	Use of DC75 does not require a license.
DCIO	None	Use of DCIO does not require a license.
DSP	DSP cores	See DSP Software Licensing Requirements.
DSSC	Number of PatGens enabled for DSSC	See DSSC Software Licensing Requirements.
Embedded Signal Analyzer (ESA) Toolkit	- Enable ESA Toolkit 2G cellular 3G cellular - Connectivity	See ESA Toolkit Software Licensing Requirements.

		• WiMax Digital Broadcast
		• LTE
GigaDig	None	Some license files may contain obsolete license feature lines for GigaDig. However, a license for GigaDig is not required.
GPIB	None	Use of GPIB does not require a license.
HDVS	Number of channels	See HDVS Software Licensing Requirements.
HexVS	None	Use of HexVS does not require a license.
HPM	None	Use of the High Performance Memory option does not require a license. Use of the Bitmap Tool HPM requires a license. See Bitmap Tool HPM Software Licensing Requirements.
HSD1000	• Data rate • Pattern depth • Differential pins • Enhanced accuracy	See HSD1000 Software Licensing Requirements.
Memory Test Option (MTO)	• Number of digital channels enabled for MTO • Bitmap Tool, Engineering mode	See MTO Software Licensing Requirements.

- [Bitmap Tool, Production mode](#)
- [Redundancy Analyzer](#)
- [RA/Plus algorithm](#)

Precision Metering Option (PMO)	None	Use of PMO does not require a license.
SB6G	Data rate	See SB6G Software Licensing Requirements.
SCAN	Number of digital pins in scan patterns enabled for SCAN	See SCAN Software Licensing Requirements
SECS/GEM	Enable	See SECS/GEM Software Licensing Requirements.
Support Board	Support Board features: • DIB Power • DIB and Probe Card EEPROMs • Utility Bits, Extended Utility Bits • Sync Panel	Use of the Support Board features does not require a license.
Test Head	Slots	This is the number of boards mapped in the test system, excluding support and distribution boards. In 8.20.00 and later, the number of slots is a counted feature.
TurboAC Capture	• Number of channels	Uses same license as BBAC. See TurboAC Capture Software Licensing

	• Bandwidth	Requirements.
TurboAC Source	• Number of channels • Bandwidth	Uses same license as BBAC. See TurboAC Source Software Licensing Requirements.
UltraPAC80 Capture	• Channels and bandwidth	See UltraPAC80 Capture Software Licensing Requirements.
UltraPAC80 Source	• Channels and bandwidth	See UltraPAC80 Source Software Licensing Requirements.
UltraPin4000	None	Use of UltraPin4000 does not require a license.
UltraPin800	• Data rate • Memory depth • High Voltage pin • Differential pins • Enhanced accuracy	See UltraPin800 Software Licensing Requirements.
UltraPin1600	• Channels • Data rate • Memory depth	See UltraPin1600 Software Licensing Requirements.

- [Digital loopback](#)
- [Protocol Aware, DDR](#)
- [Protocol Aware, nWire](#)
- [Protocol Aware, Loopback](#)
- [Protocol Aware, PRBS](#)

UltraSerial10G

See [UltraSerial10G Software Licensing Requirements](#).

- [Channels](#)
- [Data rate](#)
- [Digital loopback](#)
- [PRBS](#)
- [Jitter Measure](#)
- [Jitter Insert](#)
- [Protocol Aware, PCIe End Point](#)
- [Protocol Aware, PCIe Root Complex](#)
- [Protocol Aware, sWire](#)

UltraVI80

DCVI channels

See [UltraVI80 Software Licensing Requirements](#).

UltraWave 12G	- Arbitrary Waveform Generator - Noise Source - Number of channels - Phase Noise Analyzer - Source Amplitude Modules - Vector Network Analyzer	See UltraWave 12G Software Licensing Requirements.
UltraWave24	- Arbitrary Waveform Generator - Noise Source - Number of channels - Receiver bandwidth - Source Amplitude Modules - Vector Network Analyzer	See UltraWave24 Software Licensing Requirements.
UVS256	None	Use of UltraVS256 does not require a license.
VHFAC	- Capture - Capture sample rate	See VHFAC Software Licensing Requirements.

- [Source](#)
- [Source sample rate](#)

VSM

None

Use of the Voltage Supply Module (VSM) does not require a license.

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IG-XL Licensing : Available UltraFLEX Features

Available UltraFLEX Features

The following feature names may appear in a license for an UltraFLEX test system.

IG-XL Instrument and Software Licensing for UltraFLEX

Feature Name	Instrument	Feature	Comments
8UniversalRFPorts	UltraWave 12G	Universal 8 Port License	Counted license. Each license enables eight ports on one RF measurement board.
16UniversalRFPorts	UltraWave 12G	Universal 16 (13-16) Port License	Counted license. Each license enables 16 ports on one RF measurement board.
12GHzReceiver	UltraWave 12G	12GHz Receiver	Counted license. One license required per receiver operating at 12GHz (default performance is 6GHz).
12GHzSource	UltraWave 12G	12GHz Source	System wide license. If enabled, this license allows 12GHz on all RF ports in the test system.

1PortVNA	UltraWave 12G	Universal Port VNA Option 1 port	Counted per measurement board. Allows 1-port VNA measurement on any port on the MWMMeasure Board.
BBACCap	BBAC Capture	Number of instruments (channels)	See BBAC Capture Software Licensing Requirements.
BBACCapBandwidth	BBAC Capture	Bandwidth	
BBACSrc	BBAC Source	Number of instruments (channels)	See BBAC Source Software Licensing Requirements.
BBACSrcBandwidth	BBAC Source	Bandwidth	
BitmapTool.Integrated	HPM	Bitmap Tool (Production)	See Bitmap Tool HPM Software Licensing Requirements.
BitmapTool.StandAlone	HPM	Bitmap Tool (Engineering)	
CTExpert	DataTool	CTExpert Tool	Counted license. Required to use the CTExpert Tool (standalone tool).
DC30	DC30	Enable	Obsolete
DC75	DC75	Enable	Obsolete
DDRMemoryEngine	UltraPin1600	DDR Memory Protocol Aware	Counted license. Each DDRMemory engine requires one license, up to 4 engines per board.
DSPPCConcurrentAlgCount	DSP Service	Number of concurrent worker threads run on a DSP system	See DSP Software Licensing Requirements.

DUTLOSource	UltraWave 12G	DUT LO Source	Counted license. One license required per MWSynthRef board to use the DUT low source.
HDVS1Channels	HDVS	Number of channels	See HDVS Software Licensing Requirements.
HSD_DSSC	HSD1000, UltraPin800, UltraPin1600	Per-board Digital Signal Source and Capture	See DSSC Software Licensing Requirements.
HSD_MemoryDepth	HSD1000, UltraPin800, UltraPin1600	Memory depth	HSD1000 Software Licensing Requirements UltraPin800 Software Licensing Requirements UltraPin1600 Software Licensing Requirements
HSD_MTO	HSD1000, UltraPin800, UltraPin1600	Per-board Memory Test Option	See MTO Software Licensing Requirements.
HSD_Scan	HSD1000, UltraPin800, UltraPin1600	Per-board SCAN	See SCAN Software Licensing Requirements.
HSD_Speed	HSD1000, UltraPin800, UltraPin1600	Maximum data rate	HSD1000 Software Licensing Requirements UltraPin800 Software Licensing Requirements UltraPin1600 Software Licensing Requirements

HSD1600_Channels	UltraPin1600	Number of channels	See UltraPin1600 Software Licensing Requirements
HSS_Speed	SB6G	Maximum data rate	See SB6G Software Licensing Requirements.
HSS10G_Channels	UltraSerial10G	Number of channel cards	See UltraSerial10G Software Licensing Requirements.
HSS10G_JitterInsert	UltraSerial10G	Jitter Insertion	See UltraSerial10G Software Licensing Requirements.
HSS10G_JitterMeasure	UltraSerial10G	Jitter Measurement	See UltraSerial10G Software Licensing Requirements.
HSS10G_Loopback	UltraSerial10G	Digital Loopback Protocol Aware	See UltraSerial10G Software Licensing Requirements.
HSS10G_PRBS	UltraSerial10G	PRBS Protocol Aware	See UltraSerial10G Software Licensing Requirements.
HSS10G_Speed	UltraSerial10G	Maximum data rate	See UltraSerial10G Software Licensing Requirements.
IG-XL_Bitmap	MTO	Bitmap Tool enable	See MTO Software Licensing Requirements.
IGXLConcurrentExecution	DataTool	Concurrent Test, Asynchronous pattern start	Counted license. Required to use concurrent opcode or asynchronous pattern starts. See Concurrent Test Software Licensing Requirements .

IG-XL_DataTool	DataTool	Enable	Obsolete
IG-XL_DataTool_Jaguar	DataTool	Enable	
IG-XL_HAND_SWITCH	IG-XL	Start/Stop hand control	
IG-XL_PBM	MTO	Production Bitmap Tool enable	See MTO Software Licensing Requirements.
IG-XL_RA	MTO	Redundancy Analysis	See MTO Software Licensing Requirements.
IG-XL_RAPlus	MTO	RA/Plus algorithm	See MTO Software Licensing Requirements.
IG-XL_SECSGEM	SECS/GEM	Enable	
IG-XL_TesterServiceDaemon	DataTool	Test cell integration	
IG-XL_Wafermap	Wafer Map Tool	Enable	
LoopbackEngine	UltraPin1600	Digital Loopback Protocol Aware	Counted license. Each digital loopback engine requires one license, up to 64 engines per board.
LowPhaseNoiseDUTRefSource	UltraWave 12G	Low Phase Noise DUT/Clock Reference Source	Counted license. One license required per MWSynthRef board to use the low phase noise DUT/Clock reference source.
MainSlots	Test head slots	Enable	In 8.20.00 and later, if a NumSlots license is not present, IG-XL checks for MainSlots.

NoiseSource	UltraWave 12G	On-Board Source Noise	Counted license. One license required per measurement board.
NumSlots	Test Head	Slots	Counted license in 8.20.00 and later. One license required per test system. IG-XL checks for a MainSlots license if NumSlots is not available.
nWireEngine	UltraPin1600	nWire Protocol Aware	Counted license. Each nWire engine requires one license, up to 16 engines per board.
OnBoardDUALAWGSynthDR	UltraWave 12G	On-Board Dual AWG (Synthesizer DR or LO)	Counted license. One license required per dual AWG board for modulated source.
PCleEndPointEngine	UltraSerial10G	PCle Protocol Aware	Counted license. Each PCle end point engine requires one license, up to 20 engines per board.
PCleRootComplexEngine	UltraSerial10G	PCle Protocol Aware	Counted license. Each PCle root complex engine requires one license, up to 20 engines per board.
PRBSEngine	UltraPin1600	PRBS Protocol Aware	Counted license. Each PRBS engine requires one license, up to 64 engines per board.
Scan	HSD1000	SCAN testing	Obsolete
ScanPins	HSD1000	SCAN testing	Obsolete

sWireEngine	UltraSerial10G	sWire Protocol Aware	Counted license. Each sWire engine requires one license, up to 20 engines per board.
TimeLines	DataTool	TimeLines Tool	Counted license. Required to use the TimeLines Tool (standalone tool).
UltraCapture	UltraPAC80	Bandwidth	Counted license. One license required per UPAC80 capture channel. See UltraPAC80 Capture Software Licensing Requirements.
UltraSource	UltraPAC80	Bandwidth	Counted license. One license required per UPAC80 source channel. See UltraPAC80 Source Software Licensing Requirements.
UltraVChannel	UltraVI80	DCVI channel	Counted license. One license required per UltraVI80 DCVI channel. See UltraVI80 Software Licensing Requirements.
UW24_1PortVNA1	UltraWave24	UltraWave24 1 Port VNA	Counted license. One license required per UltraWave24 measure board. See UltraWave24 Software Licensing Requirements.

UW24_AWG	UltraWave24	UltraWave24 Modulated Source, 80 MHz and 160 MHz Bandwidths	Counted license. One license required per UltraWave24 modulated source. See UltraWave24 Software Licensing Requirements.
UW24_DualAWG1	UltraWave24	UltraWave24 Dual AWG	Counted license. See UltraWave24 Software Licensing Requirements.
UW24_DualSynthesizer1	UltraWave24	UltraWave24 Dual Synthesizer	Counted license. See UltraWave24 Software Licensing Requirements.
UW24_DutRef1	UltraWave24	UltraWave24 DUT Reference	Counted license. One license required per UltraWave24 DUT reference. See UltraWave24 Software Licensing Requirements.
UW24_EnMeas1	UltraWave24	UltraWave24 Enhanced Measurement	Counted license. See UltraWave24 Software Licensing Requirements.
UW24_EnMultiTone1	UltraWave24	UltraWave24 Enhanced Multitone	Counted license. See UltraWave24 Software Licensing Requirements.
UW24_NoiseSource1	UltraWave24	UltraWave24 Noise Source	Counted license. One license required per UltraWave24 measure board. See UltraWave24 Software

			Licensing Requirements.
UW24_Ports1	UltraWave24	UltraWave24 RF Port	Counted license. One license required per UltraWave24 port. See UltraWave24 Software Licensing Requirements.
UW24_RcvEnable1	UltraWave24	UltraWave24 Receiver Enable	Counted license. One license required per UltraWave24 measure board. See UltraWave24 Software Licensing Requirements.
UW24_RcvFrequency700	UltraWave24	UltraWave24 Receiver Frequency > 700 MHz	Counted license. One license required per UltraWave24 measure board. See UltraWave24 Software Licensing Requirements.
UW24_Receivers	UltraWave24	UltraWave24 Receiver, 45 MHz, 100 MHz, and 160 MHz Bandwidth	Counted license. One license required per UltraWave24 receiver. See UltraWave24 Software Licensing Requirements.
UW24_SourceFrequency700	UltraWave24	UltraWave24 Source Frequency > 700 MHz	Counted license. One license required per UltraWave24 source board. See UltraWave24 Software Licensing Requirements.

UW24_StdMeas1	UltraWave24	UltraWave24 Standard Measurement	Counted license. One license required per UltraWave24 measure board. See UltraWave24 Software Licensing Requirements.
UW24_StdMultiTone1	UltraWave24	UltraWave24 Standard Multitone	Counted license. One license required per UltraWave24 measure board. See UltraWave24 Software Licensing Requirements.
VHFACCapture	VHFAC	Number of channels	See VHFAC Software Licensing Requirements.
VHFACCaptureSamplingRate	VHFAC	Sampling rate	
VHFACSource	VHFAC	Number of channels	
VHFACSourceSamplingRate	VHFAC	Sampling rate	

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Remote Test System Connections

Remote Test System Connections

Using Remote Desktop Connection with Windows 7

You can use Microsoft’s Remote Desktop Connection (RDC) in Windows 7 to make remote connections to test systems while working online.

Unlike RDC in Windows XP, RDC in Windows 7 does not terminate an existing local session when another person remotely logs in using RDC. Instead, RDC preserves the local session and puts it into a disconnected state. This change in handling local and remote sessions can create problems if more than one person is logged on to the computer (for example, when one user is running IG-XL while another user is installing IG-XL, or both users are running IG-XL simultaneously).

As an alternative to RDC, you can also use VNC to connect to the test system remotely.

To ensure that all IG-XL tools and utilities operate properly (for example, DataCollect’s OutputWin window), you must change the default RDC connection options.

- | | |
|----|---|
| 1. | Open RDC and click the Options button to open the RDC options tabs. |
| 2. | Click the Local Resources tab and clear the Printers option in the local devices and resources section. |
| 3. | Click the More button and clear the Smart cards, Ports, Drives, and Other supported Plug and Play (PnP) devices options in this window. |

Using a VNC Connection

You can also use VNC to make remote connections to test systems while working online. When running IG-XL on a test system that you are accessing through VNC, you must set certain VNC settings to ensure that certain aspects of the IG-XL tools function correctly.

- [Enable](#) Capture Alpha-Blending on the VNC server.

IG-XL requires this setting for tooltips and other alpha-blending that the tools use, such as the “Needs refresh” window in PatternTool, HRAM Display, and Waveform Display.

- [Disable clipboard transfer by checking](#) Disable clipboard transfer in the VNC viewer settings.

This lets you use the PatternTool cut, copy, and paste operations and other operations that use cut, copy, and paste, such as creating a new subroutine.

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Validating Your Operating Environment

Validating Your Operating Environment

The System Validation utility is a stand-alone tool that inspects your current software and hardware environment and reports on whether your environment meets the requirements for the release of IG-XL you are running. This can help you to ensure that your test program runs on a test system that fully supports the resources and IG-XL version your test program expects to find.

When you run the utility offline, it reports your operating system version and the version of Microsoft Office Excel you have installed.

When you run the utility on a test system, it reports information about the tester computer hardware and operating system and the version of Excel, and the operating system and hardware of any connected DSP computers.

Both online and offline, the utility validates that the components of your system meet the minimum system requirement for the active version of IG-XL.

You can also use the tool to view the minimum system requirements for the version of IG-XL you are running.

For complete help on using the System Validation utility, start the utility and select Help>System Validity Check Help.

This section includes the following topics:

- | | |
|---|---|
| • | About System Validation |
| • | Starting the System Validation Standalone Utility |
| • | Defining System Validation Check as an IG-XL Add-In |

- [Validating the Operating Environment in a Test Program](#)

	
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Validating Your Operating Environment : About System Validation

About System Validation

Each version of IG-XL has a set of hardware and software requirements. These define the minimum configurations that IG-XL needs to run properly. Teradyne publishes these in the System Requirements section of the Release Notes for each version.

When you run the System Validation utility, it compares the current system configuration against the requirements for the active version of IG-XL.

To keep the information current, Teradyne releases updates to the System Validation utility separately from IG-XL, as well as including it as part of IG-XL installation. You can download the most recent version from eKnowledge.

The System Validation utility requires that you have a version of IG-XL for UltraFLEX installed and set as the current version. If you do not have IG-XL for UltraFLEX installed or if it is not the current version, the utility returns an error.

When you run System Validation offline, it gathers the following information:

- | | |
|---|---|
| • | User name with which you logged on to Windows |
| • | Computer name |
| • | Tester mode ("OFF-LINE") |
| • | IG-XL version |
| • | Operating system version |

- Operating system edition

- Operating system service pack level

- Operating system type (32-bit or 64-bit)

- Microsoft Office version

When you run the utility on a tester computer, it gathers the following information:

- User name with which you logged on to Windows

- Computer name

- Tester mode ("ONLINE")

- IG-XL version

- Operating system version

- Operating system edition

- Operating system service pack level

- Operating system type (32-bit or 64-bit)

- Number of processors

- Processor speed

- Processor model

- Microsoft Office version

- Computer type

- Physical memory installed, in MB

If a test system has DSP computers attached, the utility gathers the following information about each DSP computer:

- Operating system version

- Operating system edition

- Operating system service pack level

- Operating system type (32-bit or 64-bit)

- Number of processors

- Number of cores per processor

- Level1 cache size, in KB

- Level2 cache size, in KB

- Level3 cache size, in MB

- Processor speed

- Physical memory installed, in MB

- Processor type

- Processor model

- Detailed processor family and model information

- Computer type

- IP address

The utility does not record information for DSP computers if you are running offline, even if CurrentConfig.Txt contains a Virtual DSP Board.

In addition to reporting whether your system configuration passes or fails the requirements check, the utility stores the gathered information in the following text file:

C:\Users\Public\Documents\Teradyne\IG-XL\SystemValidityReport.txt

The text file includes the date and time of the report as the first line in the file.

The utility writes to the same file each time you produce a report. If you want to keep a record of a system's configuration report, rename the file after you exit the utility. This prevents future reports from overwriting the existing log.

The following system validation report shows a tester computer and two DSP computers:

System Validation Started....

```
User Name:      Programmer
Machine Name:   ULTRAFLEX1
Tester Mode:    ONLINE
IG-XL Version:  8.30.00_uflx
```

System Validation on Host PC completed Successfully.

1. OS Version: Windows 7
2. OS Edition: Embedded
3. OS ServicePack: 1
4. OS Bits: 64-Bit
5. Number of Processor: 1
6. Processor Speed: 3.30GHz
7. Processor Type: E5-2643 0
8. MSO Version: Microsoft Excel 2010 ServicePack: 1
9. Host Processor Type: Tera1Windows7
10. Physical Memory(MB): 8192

System Validation on DSPPC0 completed Successfully.

1. OS Version: Windows 7
2. OS Edition: Embedded
3. OS ServicePack: 1
4. OS Bits: 64-Bit

- 5. Number of Processor: 1
- 6. Number of Cores Per Processor: 10
- 7. L1 CacheSize(KB): 32
- 8. L2 CacheSize(KB): 256
- 9. L3 CacheSize(MB): 25
- 10. Processor Speed: 2.80GHz
- 11. Physical Memory(MB): 65536
- 12. Processor Type: E5-2680 v2
- 13. Processor Model: S2600GZ
- 14. Host Processor Type: Tera1Windows7
- 15. IP Address: 10.200.200.1

System Validation on DSPPC1 completed Successfully.

- 1. OS Version: Windows 7
- 2. OS Edition: Embedded
- 3. OS ServicePack: 1
- 4. OS Bits: 64-Bit
- 5. Number of Processor: 1
- 6. Number of Cores Per Processor: 10
- 7. L1 CacheSize(KB): 32
- 8. L2 CacheSize(KB): 256
- 9. L3 CacheSize(MB): 25
- 10. Processor Speed: 2.80GHz
- 11. Physical Memory(MB): 65536
- 12. Processor Type: E5-2680 v2
- 13. Processor Model: S2600GZ
- 14. Host Processor Type: Tera1Windows7
- 15. IP Address: 10.200.200.254

System Validation: PASSED

Log Report is saved in c:\Users\Public\Documents\Teradyne\IG-XL\SystemValidityReport.txt

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Validating Your Operating Environment : Starting the System Validation Standalone Utility

Starting the System Validation Standalone Utility

You can run the system validation check through the utility's graphical interface or from the command line.

To start the external utility's graphical interface:

1. In Windows Explorer, browse to the following folder:

C:\Program Files (x86)\Teradyne\IG-XL\<version>\bin

where <version> is the version of IG-XL you are running.

This path assumes that you installed IG-XL in the recommended folder. If you installed IG-XL in a different folder, adjust the path according to your installation.

2. Double-click ValidateSystem.Exe.

The utility opens and performs the validity check.

To run the external utility from the command line:

1. Open a Command Prompt.

Select Start>All Programs>Accessories>Command Prompt.

2. Change the current directory to the bin directory of the current IG-XL installation.

If you installed IG-XL in the recommended folder, type the following command:

CD C:\Program Files (x86)\Teradyne\IG-XL\<version>\bin

where <version> is the version of IG-XL you are running.

If you installed IG-XL in a different folder, adjust the path according to your installation.

3. Type ValidateSystem r and press Enter.

The utility displays information about your operating environment in the Command window. Note that you cannot run the utility from the command line if the graphical version is running. For information on defining the utility as an add-in so you can launch it from within IG-XL, see [Defining System Validation Check as an IG-XL Add-In](#). For complete help on using the System Validation utility, start the utility and select [Help>System Validity Check Help](#).

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Validating Your Operating Environment : Defining System Validation Check as an IG-XL Add-In

Defining System Validation Check as an IG-XL Add-In

If you want to be able to call the system validation check utility from within IG-XL, you can define the program as an add-in. To do this:

- | | |
|----|---|
| 1. | From the Excel menu bar, select the IG-XL Tools ribbon. |
| 2. | Select Add-Ins>Add-Ins Settings. |

IG-XL Settings opens to the Add-Ins page.

- | | |
|----|---|
| 3. | In the Executable name field, type System Validity Check. |
| 4. | In the Executable file location field, type the path to the program file. |

If you installed IG-XL in the recommended folder, this is:
C:\Program Files (x86)\Teradyne\IG-XL\8.30.00_uflx\bin\ValidateSystem.Exe
If you installed IG-XL in a different folder, adjust the path according to your installation.

- | | |
|----|-----------|
| 5. | Click OK. |
|----|-----------|

IG-XL Settings adds System Validity Check to the list.
Once you define System Validation as an add-in, you can launch it by selecting Add-Ins>System Validity Check from the IG-XL Tools ribbon.
For more information on creating and managing the list of add-ins, see Add Ins, Tool Sets, and IG-XL Settings in DataTool help.

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Validating Your Operating Environment : Validating the Operating Environment in a Test Program

Validating the Operating Environment in a Test Program

Checking the validity of your operating environment from within your test program can help ensure that your test program runs in the environment that properly supports the resources and features it requires.

To run the validity check within a test program, call `TheExec.ValidateSystem`.

Related Topics

- [TheExec.ValidateSystem](#)

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Switching IG-XL Versions

Switching IG-XL Versions

IG-XL Version Selector Plus (VS+) is a standalone utility that lets you switch between versions of IG-XL. If you have multiple versions of IG-XL installed, you can switch between versions without uninstalling or reinstalling. Using VS+, you can apply or roll back any installed patches within a release, switch to a specific patch of a different release, or switch among base versions of any installed release. VS+ handles any dependencies among patches. This can help you create exactly the same IG-XL environment on multiple test systems.

You can start this utility either from the Windows Start menu or from a command line. VS+ unregisters the current version of IG-XL and registers the new selected version.

VS+ supports releases of IG-XL for the J750, FLEX, and UltraFLEX platforms.

VS+ is not distributed with new IG-XL releases. The IG-XL installation adds a link to download the latest version of VS+. You can download VS+ directly from Teradyne’s eKnowledge portal. To allow it to manage all versions of IG-XL, VS+ releases are not tied to any specific IG-XL release.

For more information and instructions on using VS+, install the tool, then open the included help. To install VS+, use Start>All Programs>Teradyne Version Selector Plus for directions and a link to eKnowledge.

If you have already installed the program, use this menu item to start the program.

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Virus Protection

Virus Protection

Note the following regarding virus protection and security on Teradyne test systems:

- Teradyne policy is to protect customer workstations in our production line.
 - Our workstation PCs are purchased from an integrator without attachment to any existing network.
 - All software loaded on the workstation by the integrator comes directly from CDs provided by Teradyne.
- When test systems reach our final configuration step, we install Symantec antivirus software and scan the workstation before connection to our in-house network.
 - Symantec software remains loaded until all final configuration and test steps are complete.
 - Symantec antivirus software is removed from the workstation before shipment to customers.
- Teradyne customers use many types of malware protection software.
 - Software selection and security processing is often determined by local customer IT departments.

- Teradyne does not endorse a specific malware security software brand.
- We do recommend use of a “major provider” software product.
- Install and use antivirus software on any test system.

Load antivirus software before connection to a network or to the Internet.

- Teradyne does not recommend installing antivirus software on the DSP PC. Such software is not required because there is:
 - No user access to the DSP PC
 - No network access to the DSP PC

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